Lecture 16
Markov Chain Monte Carlo Algorithms

Algorithms, AI & Data Science II

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Motivation

- we introduced Markov models of stochastic processes
- we explored under which conditions Markov chains reach a **unique stationary state**
- stationary states are basis for important **algorithmic applications of Markov chains**
- we can use stationary state of **random surfer in Web graph** to rank Web pages based on **relevance or importance**
- we can use Markov chains for the **heuristic exploration of large state spaces**

DALL-E image generated for prompt "searching a needle in a haystack"

image credit: www.bing.com

Notes:

- **Lecture 16: Markov Chain Monte Carlo Algorithms** 02.07.2025
- **Educational objective:** We explore Markov Chain Monte Carlo algorithms. We show how Markov chains can be used to sample from arbitrary distributions, and how we can use physics-inspired simulated annealing to address NP-hard optimization problems.
  - Markov Chain Monte Carlo
  - Metropolis Sampling
  - Optimization by Simulated Annealing
- **Handout version:** *2025/06/24 14:49:50*

Notes:

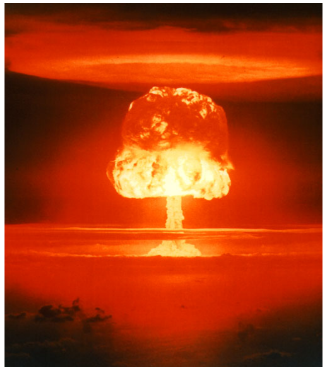
Dealing with large state spaces

- we often face algorithmic problems in **huge state spaces** that are too large to enumerate
  - NP-hard optimization problems
  - sampling from state spaces with complex distributions

application: computational physics

- statistical physics studies **macroscopic properties of materials** comprised of large number of interacting particles
- probabilities of **specific particle arrangements** depend on temperature, energy, etc.
- **huge space** of possible arrangements of a very large number of particles (e.g. all possible positions/velocities of $\approx 10^{23}$ atoms)

- how can we **efficiently sample from huge state spaces**?

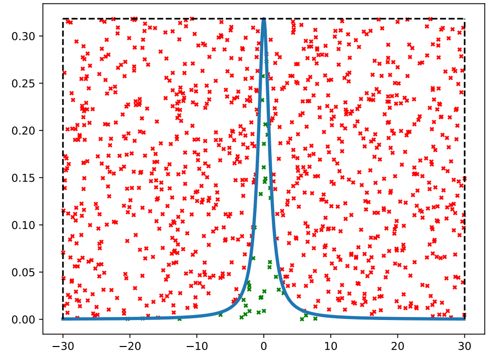


Castle Romeo test of a hydrogen bomb

image credit: United States Department of Energy, public domain

Rejection sampling?

- in L09, we introduced **rejection sampling algorithm**
- can be used to sample from **arbitrary (continuous) distributions**
- is this **efficient** for any distribution?
- consider complex distribution where **probabilities are concentrated in small portions of state space**
- we **reject huge number of samples**
- can we **use Markov chains** to derive a more efficient sampling technique?



accepted (green) and rejected (red) samples generated by **rejection sampling algorithm**

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Markov Chain Monte Carlo

- for given transition matrix $\mathbf{T}$ of Markov chain, **we can calculate stationary distribution π of states**
- idea: define Markov model such that for $t \to \infty$ state probabilities converge to **desired unique target distribution π**

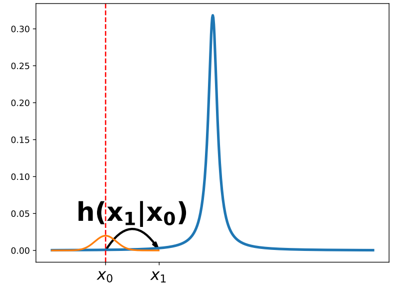
general approach

- start from random initial state
- simulate random walk through state space
- consider state that we arrive at after sufficiently large number of steps

- basis for **Markov Chain Monte Carlo algorithms** widely used in data science, statistics, computational physics, or bioinformatics

Casino of Monte Carlo

Metropolis sampling



- consider that we know function $f(x)$ and want to sample from **target distribution π with $\pi_x \propto f(x)$** for states x
- we pick a random initial state x_0
- assuming that we are in state x_t, consider iterative algorithm that samples new state x_{t+1} from **symmetric proposal distribution** $g(x_{t+1}|x_t) = g(x_t|x_{t+1})$
- we accept new state x_{t+1} with probability

$$h(x_{t+1}|x_t) := \min\left(1, \frac{f(x_{t+1})}{f(x_t)}\right)$$

  or stay in current state x_t
- for $t \to \infty$ **Metropolis algorithm samples states x with probabilities π_x**

Metropolis algorithm → N Metropolis et al., 1953

procedure METROPOLIS_SAMPLING(f, g, N)
  $x \leftarrow initial\_state()$
  for i in range(N) **do**
    $x\_new \leftarrow sample(g, x)$
    $p \leftarrow uniform(0, 1)$
    if $p \leq \min(1, f(x\_new)/f(x))$ **then**
      $x \leftarrow x\_new$
    end if
  end for
  return x
end procedure

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Metropolis sampling as Markov chain

- for a discrete state space, let us consider the **Markov chain defined by Metropolis algorithm**
- since the generation and acceptance of proposal are independent, we obtain the following **entries of the transition matrix $\mathbf{T}$** of the Markov chain

$$T_{x_t, x_{t+1}} = g(x_{t+1}|x_t) \cdot \min\left(1, \frac{f(x_{t+1})}{f(x_t)}\right) \qquad T_{x_{t+1}, x_t} = g(x_t|x_{t+1}) \min\left(1, \frac{f(x_t)}{f(x_{t+1})}\right)$$

- since the **proposal distribution is symmetric**, i.e. $g(x_{t+1}|x_t) = g(x_t|x_{t+1}) =: \epsilon$, we have

$$T_{x_t, x_{t+1}} = \epsilon \cdot \min\left(1, \frac{f(x_{t+1})}{f(x_t)}\right) \qquad T_{x_{t+1}, x_t} = \epsilon \cdot \min\left(1, \frac{f(x_t)}{f(x_{t+1})}\right)$$

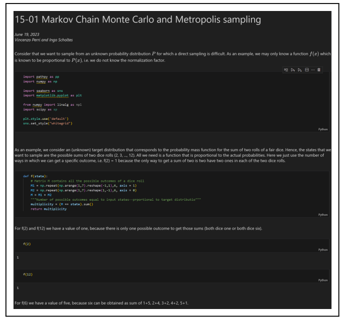
- for states x_t and x_{t+1} with $f(x_{t+1}) > f(x_t)$ we then obtain

$$T_{x_t, x_{t+1}} = \epsilon \qquad \text{and} \qquad T_{x_{t+1}, x_t} = \epsilon \cdot \frac{f(x_t)}{f(x_{t+1})} < \epsilon$$

i.e. ϵ is probability to move to state x_{t+1} with "larger" value $f(x_{t+1})$

Practice Session

- we use the Metropolis sampling algorithm to sample from the probability mass function of a random variable X assuming the sum of two rolls of a fair dice
- we use a random walk in an arbitrary connected graph to generate a proposal distribution
- as function $f(x)$ we use the number of possible combinations in which can obtain a sum of x
- we calculate the stationary state of the resulting Markov chain and show that it converges towards to desired target distribution



practice session

see notebook 16-01 in `gitlab` repository at

→ https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2

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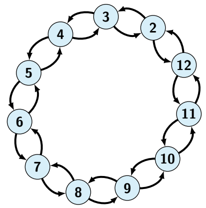
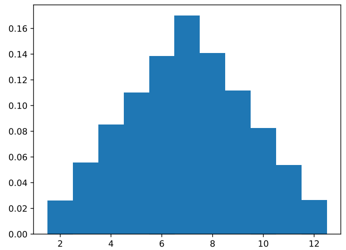
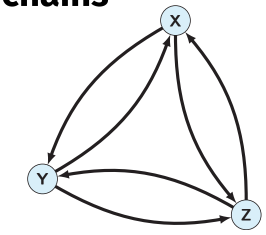
Analysis of Metropolis sampling



- for our simple example, we experimentally confirmed that the Metropolis sampling algorithm converges to "correct" stationary state π with $\pi_i \propto f(i)$
- but **why** does this actually work?
- we know that a unique stationary state exists if Markov chain is **irreducible and aperiodic** $\rightarrow$ L15
- need to show that target distribution π is **stationary state of Markov chain** defined by Metropolis algorithm
- proof idea: symmetry of proposal distribution allows us to show a strong property that yields **sufficient condition for stationary state**



Detailed balance and reversible Markov chains



- we say that Markov chain with transition matrix $\mathbf{T}$ is **reversible** (satisfies the **detailed balance condition**) iff a stationary state π exists such that

$$\forall i, j : \pi_i T_{ij} = \pi_j T_{ji}$$

- for a reversible Markov chain, we can **reverse time** and obtain the same stochastic process
- detailed balance holds for $\mathbf{T}$ and $\pi \implies \pi$ *stationary state* for $\mathbf{T}$
- π stationary state for $\mathbf{T} \not\Longrightarrow$ detailed balance holds for $\mathbf{T}$ and π
- i.e. detailed balance condition is **sufficient but not necessary condition** for π being stationary state of $\mathbf{T}$

$\rightarrow$ exercise sheet

$$\mathbf{T} = \begin{array}{c c} & \begin{array}{ccc} x & y & z \end{array} \\ \begin{array}{c} x \\ y \\ z \end{array} & \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} \end{array}$$

example

for $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and any pair i, j we have

$$\pi_i \cdot T_{ij} = \frac{1}{3} \cdot \frac{1}{2} = \pi_j \cdot T_{ji}$$

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Group Exercise 16-01

- Show that the Markov chain defined by the Metropolis algorithm satisfies the detailed balance condition for a stationary state π with $\pi_i \propto f(i)$.

Group Exercise 16-01

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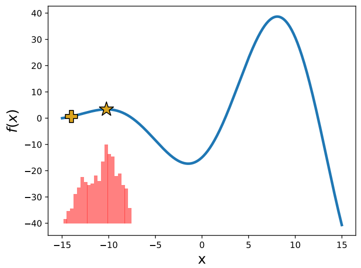
From sampling to heuristic optimization

- for given function $f(x)$, **we can define Markov chain** whose stationary state is proportional to $f(x)$
- can we define a Markov chain whose stationary state corresponds to **maximum value of $f(x)$**?
- basis for **metaheuristic optimization algorithms**
- for given $f(x)$ consider distribution of states sampled by Metropolis algorithm for **finite number of steps**
- since we are more likely to move to states with larger $f(x)$ we are **likely to be "stuck" in local maxima**
- how can we **explore the state space** and still converge to an optimum?



distribution of sampled states of **Metropolis Markov Chain Monte Carlo** for $N = 10000$ steps
(initial and maximum state marked)

Markov chains in statistical physics

- statistical physics studies how **macroscopic properties of aggregate matter** can be understood based on statistics of underlying particle arrangements

examples
- properties of water depending on temperature and pressure
- melting point of iron

- **state of the system** is described by specific configuration of particles (i.e. atoms or molecules)
- random fluctuations depend on **energy of states** and **temperature of the system**
- particles are memoryless, i.e. **transitions between states** can be modelled using **Markov chain**

Eruption of Castel Geyser in Yellowstone National Park

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A statistical physics perspective on optimization

- **Boltzmann distribution** describes how probabilities of particle arrangements x depend on their **energy** $E(x)$ **and temperature** T

$$P(x) \propto e^{-\frac{E(x)}{T}}$$

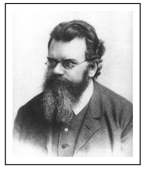
- transition probability from x to y given by **Boltzmann ratio**

$$\frac{P(y)}{P(x)} = e^{-\frac{E(y)-E(x)}{T}}$$

observations

- for $T \to \infty$ we have $e^{-\frac{E(y)-E(x)}{T}} \to e^0 = 1$
- for $T \to 0$ we have $e^{-\frac{E(y)-E(x)}{T}} \to \infty$ for $E(x) > E(y)$ and 0 otherwise

- at low temperature systems tend to move to states with **lower energy**



Ludwig Boltzmann, 1844 – 1906

image credit: public domain

blacksmith annealing metal

image credit: Jeff Kubina from Columbia, Maryland, CC-BY-SA

Optimization by simulated annealing

- transitions based on Boltzmann ratio assign high probabilities to states with **lowest energy**
- to find **maximum of function** f we can simulate "annealing", i.e. we accept state x_{t+1} with probability

$$h(x_{t+1}|x_t) = \min\left(1, e^{-\frac{f(x_t)-f(x_{t+1})}{T}}\right)$$

- we swap $f(x_t)$ and $f(x_{t+1})$ as we are interested in **maximum of** f (rather than minimum of energy)
- to explore state space and escape local maxima, we start algorithm with **high temperature**
- **cooling schedule** $T \to 0$ allows Markov chain to "settle" in states with maximum $f(x)$
- we call this algorithm **simulated annealing**

Scott Kirkpatrick

image credit: https://www.cs.huji.ac.il/w-kirk/

simulated annealing algorithm → S Kirkpatrick et al., 1983

procedure SIMULATED_ANNEALING(f, g, N, T)
  $x \leftarrow initial\_state()$
  for i in range(N) **do**
    $x\_new \leftarrow sample(g, x)$
    $p \leftarrow uniform(0, 1)$
    if $p \leq \min\left(1, e^{-\frac{f(x_t)-f(x_{t+1})}{T[i]}}\right)$ **then**
      $x \leftarrow x\_new$
    end if
  end for
end procedure

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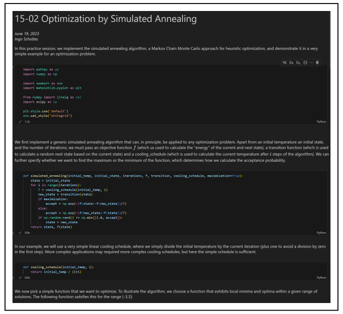
Practice Session

- we implement the simulated annealing algorithm in `python`
- we use it to address a simple optimization problem
- we explore the role of the temperature parameter in simulated annealing



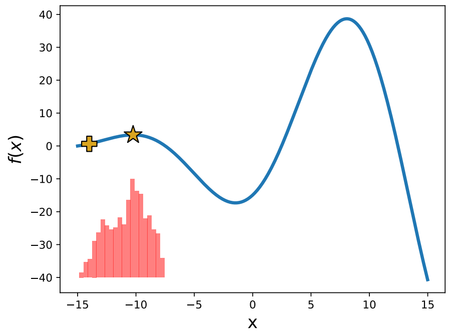
practice session

see notebook 16-02 in `gitlab` repository at

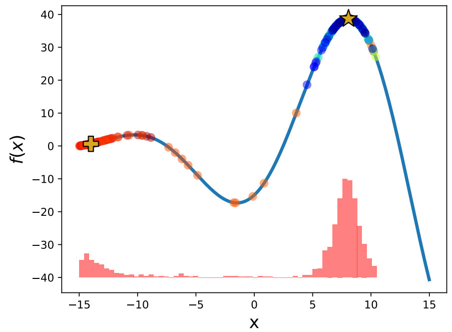
→ `https://gitlab.informatik.uni-wuerzburg.de/ml4nets_notebooks/2025_sose_akids2`

Notes:

Example



distribution of sampled states of **Metropolis Markov Chain Monte Carlo** for $N = 10000$ steps (initial and maximum state marked)

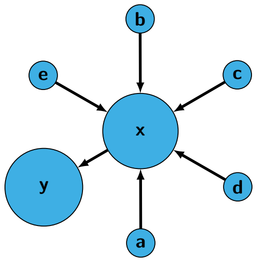


distribution of sampled states of **simulated annealing algorithm** for $N = 10000$ steps (color represents temperature T)

Notes:

In summary …

- Markov chains are theoretical basis for **algorithms for heuristic state space exploration**
- **Markov Chain Monte Carlo** is important approach in data science, statistics, computational physics, and bioinformatics
- we can use Metropolis algorithm to **sample from arbitrary distributions** (e.g. for Bayesian inference)
- Markov chain Monte Carlo is basis for **simulated annealing**
- important class of **heuristic optimization algorithms** inspired by thermodynamics
- highlights relations between **statistical physics and combinatorial optimization**



PageRank score of nodes for $\alpha = 0.85$

blacksmith forging metal

Notes:

Self-study questions

1. What is the role of the function f in Metropolis sampling?
2. What conditions have to be fulfilled for the proposal distribution $g(y|x)$ in the Metropolis algorithm?
3. Investigate the Metropolis-Hastings algorithm, a generalization of the Metropolis algorithm. How is it different from the algorithm introduced in this lecture?
4. Investigate how Metropolis sampling can be used to approximate posterior distributions in Bayesian inference.
5. What is the detailed balance condition for a Markov chain and how does it relate to the existence of a stationary state?
6. Give an example for a Markov chain with a unique stationary state that does not satisfy the detailed balance condition.
7. Show that a Markov chain with a symmetric transition matrix fulfills the detailed balance condition.
8. Give an example for a graph, where the transition matrix of a random walk is symmetric.
9. What is the Boltzmann factor and how can it be applied in heuristic optimization algorithms?
10. Explain the role of the temperature parameter in the simulated annealing algorithm.

Notes:

References

reading list

- N Metropolis, AE Rosenbluth, MN Rosenbluth, AH Teller, E Teller: **Equation of state calculations by fast computing machines**, J Chem Phys, 1953
- WK Hastings: **Monte Carlo Sampling Methods Using Markov Chains and Their Applications**, Biometrika, Vol. 57, 1970
- S Kirkpatrick, CD Gelatt Jr, MP Vecchi: **Optimization by Simulated Annealing**, Science, 1983
- O Häggström: **Finite Markov Chains and Algorithmic Applications**, 2002
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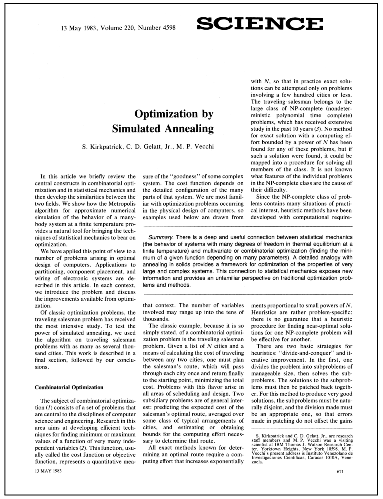
- RW Shonkwiler, F Mendivil: **Explorations in Monte Carlo Methods**, 2009



13 May 1983, Volume 220, Number 4598 **SCIENCE**

Optimization by Simulated Annealing

S. Kirkpatrick, C. D. Gelatt, Jr., M. P. Vecchi

Summary. There is a deep and useful connection between statistical mechanics (the behavior of systems with many degrees of freedom in thermal equilibrium at a finite temperature) and multivariate or combinatorial optimization (finding the minimum of a given function depending on many parameters). A detailed analogy with annealing in solids provides a framework for optimization of the properties of very large and complex systems. This connection to statistical mechanics exposes new information and provides an unfamiliar perspective on traditional optimization problems and methods.

In this article we briefly review the central constructs in combinatorial optimization and in statistical mechanics and then develop the similarities between the two fields. We show how the Metropolis algorithm for approximate numerical simulation of the behavior of a many-body system at a finite temperature provides a natural tool for bringing the techniques of statistical mechanics to bear on optimization.

We have applied this point of view to a number of problems arising in optimal design of computers. Applications to partitioning, component placement, and wiring of electronic systems are described in this article. In each context, we introduce the problem and discuss the improvements available from optimization.

Of classic optimization problems, the traveling salesman problem has received the most intensive study. To test the power of simulated annealing, we used the algorithm on traveling salesman problems with as many as several thousand cities. This work is described in a final section, followed by our conclusions.

Combinatorial Optimization

The subject of combinatorial optimization (*1*) consists of a set of problems that are central to the disciplines of computer science and engineering. Research in this area aims at developing efficient techniques for finding minimum or maximum values of a function of very many independent variables (*2*). This function, usually called the cost function or objective function, represents a quantitative measure of the "goodness" of some complex system. The cost function depends on the detailed configuration of the many parts of that system. We are most familiar with optimization problems occurring in the physical design of computers, so examples used below are drawn from that context. The number of variables involved may range up into the tens of thousands.

The classic example, because it is so simply stated, of a combinatorial optimization problem is the traveling salesman problem. Given a list of N cities and a means of calculating the cost of traveling between any two cities, one must plan the salesman's route, which will pass through each city once and return finally to the starting point, minimizing the total cost. Problems with this flavor arise in all areas of scheduling and design. Two subsidiary problems are of general interest: predicting the expected cost of the salesman's optimal route, averaged over some class of typical arrangements of cities, and estimating or obtaining bounds for the computing effort necessary to determine that route.

All exact methods known for determining an optimal route require a computing effort that increases exponentially with N, so that in practice exact solutions can be attempted only on problems involving a few hundred cities or less. The traveling salesman belongs to the large class of NP-complete (nondeterministic polynomial time complete) problems, which has received extensive study in the past 10 years (*3*). No method for exact solution with a computing effort bounded by a power of N has been found for any of these problems, but if such a solution were found, it could be mapped into a procedure for solving all members of the class. It is not known what features of the individual problems in the NP-complete class are the cause of their difficulty.

Since the NP-complete class of problems contains many situations of practical interest, heuristic methods have been developed with computational requirements proportional to small powers of N. Heuristics are rather problem-specific: there is no guarantee that a heuristic procedure for finding near-optimal solutions for one NP-complete problem will be effective for another.

There are two basic strategies for heuristics: "divide-and-conquer" and iterative improvement. In the first, one divides the problem into subproblems of manageable size, then solves the subproblems. The solutions to the subproblems must then be patched back together. For this method to produce very good solutions, the subproblems must be naturally disjoint, and the division made must be an appropriate one, so that errors made in patching do not offset the gains

S. Kirkpatrick and C. D. Gelatt, Jr., are research staff members and M. P. Vecchi was a visiting scientist at IBM Thomas J. Watson Research Center, Yorktown Heights, New York 10598. M. P. Vecchi's present address is Instituto Venezolano de Investigaciones Cientificas, Caracas 1010A, Venezuela.

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